Workflow Refinery

A human-guided compute constellation: concept paper

Dots Workflow Refinery Demo

October 2026 – Public synthetic concept

Abstract

This paper proposes an opt-in refinement layer for selected agent-assisted workflows. A human owner chooses a bounded task, reviews and redacts examples, curates a golden dataset, compares DSPy program and smaller-model candidates, and approves promotion with rollback. The aim is to test whether narrower implementations can reduce compute cost while maintaining or improving delivery quality. This is a design proposal, not a deployed system or measured result. All examples are invented.

1 Thesis and boundary

Agent-assisted work can reveal repeated computational patterns. A second-layer harness could turn *some* explicitly selected patterns into evaluated, versioned programs. The strong claim that any work in any organization can be reduced to computers under agentic control is a research thesis, not a proven universal fact. Human judgment, relationships, physical work, legal obligations, and consent set important boundaries. A system should never infer permission to collect private conversations from the existence of a repeated pattern.

The proposed "compute constellation" describes a portfolio of bounded programs and models, orchestrated under human governance. It might eventually cover internal task routing, support triage, document work, or permissioned research into jobs and contracts. This demo does none of those live activities and makes no external submissions, outreach, or model calls.

2 A controlled refinement loop

1. **Select and authorize.** A workflow owner defines the task, scope, source permissions, expected outputs, exclusions, and stop conditions. Nothing is captured by default.
2. **Review and redact.** Reviewers inspect proposed examples before dataset entry. Remove identifiers, secrets, and irrelevant material; record decisions.
3. **Curate a golden dataset.** Version representative inputs and expected outcomes. Preserve provenance and labeling rules. Adjudicate disagreements, document edge cases, and maintain a held-out evaluation set that is not used to optimize candidates.
4. **Choose candidates.** Compare a baseline against DSPy program configurations and eligible smaller models. Keep model identity, prompt/program version, dataset version, and evaluation configuration together.

5. **Evaluate.** Test quality on held-out cases, error categories, calibration, latency, and total inference cost. Inspect regressions and distribution shifts. A lower-cost candidate advances only if it meets agreed quality and safety thresholds.

6. **Approve and promote.** A human reviewer signs off on evidence, scope, and release version. Promote a versioned bundle, not a mutable label.

7. **Observe and roll back.** Watch the approved task boundary and defined metrics. Restore the previous bundle if a gate fails; record the reason and feed reviewed cases back into future dataset refinement.

3 Testable hypotheses

H1: Some bounded workflows can run on smaller models with acceptable quality. Compare cost and quality under the same task definition and held-out set.
H2: Human-adjudicated golden datasets improve repeatability and error discovery. Compare agreement, coverage, and recurring failure rates across dataset versions.
H3: Versioned artifacts, approvals, and rollback improve governance. Assess reproducibility, review time, and recovery in controlled trials.

No results are asserted here. The interactive demo's 78% and 92% scores are simulated UI values, not benchmarks or savings estimates.

4 Research domains and limits

Internal orchestration could include task routing or summary generation. External research domains could include job discovery, contract mining, and consulting lead qualification. Any such deployment would need separate authorization, data rights, organizational policy, human review, and domain-specific evaluation. This paper does not authorize or perform applications, bids, closing, or outreach. "Autonomous" here names a possible future operating mode with human gates, not a present product capability.

5 Prototype scope

The public React and Storybook showcase demonstrates selection, review, simulated evaluation, promotion, and rollback in local browser state. Playwright tests the actual interaction and records a synthetic walkthrough. There is no DSPy runtime, training, private data connection, provider call, or deployed microservice. Source and artifacts: public repository.

6 Eight architecture views

The next five figures explain the substantive control and evidence mechanisms. The final three show composition and time. Solid teal denotes an observed bounded pattern or stated owner premise; dashed amber denotes a proposed component. Labels are synthetic and do not expose private source data.

6.1 Durable identity binding

A session route must be tied to a repository, host process, managed session, and message thread. Canonical worktree matching and verified process identity prevent a label alone from becoming authority. The public figure abstracts a private implementation pattern; it does not imply this service is active.

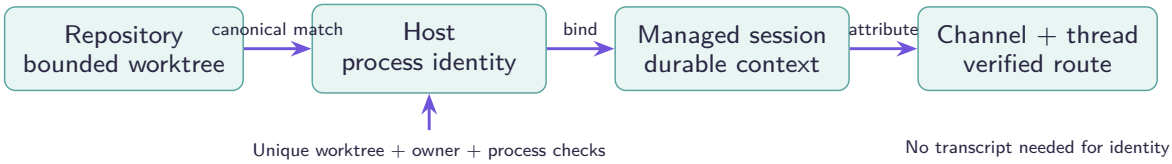


Figure 1. Attribution depends on verified identity, not transcript collection.

6.2 Managed-session lifecycle

A bounded service can launch a managed turn only through an owner-gated path and bind it after a verified session-start event. External sessions remain observation-only unless separately enrolled. Resume and stop require the bound identity.

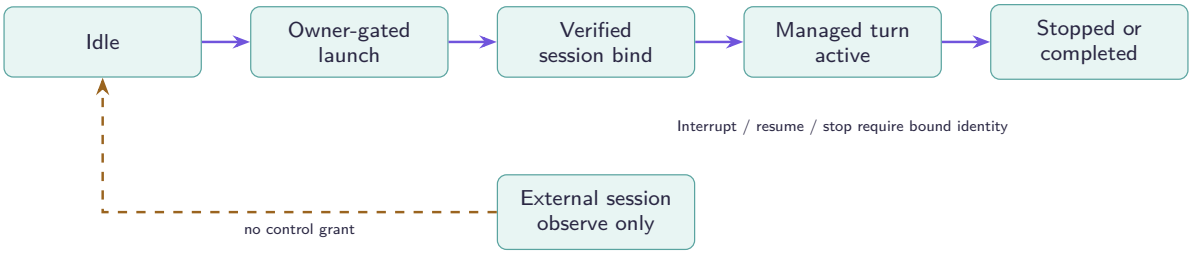


Figure 2. Managed controls and external observation remain distinct.

6.3 Reaction, approval, and execution

A reaction is a signal to inspect, not a universal grant. Actor identity, action scope, target, and limits must be checked before a reviewable plan becomes an authorized action. Uncertain results require reconciliation before retry.

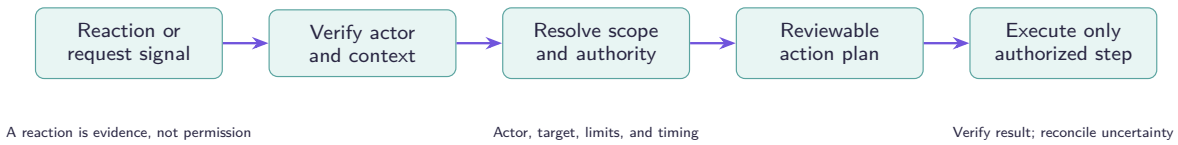


Figure 3. Evidence is separated from authority and execution.

6.4 Reviewed golden-data provenance

The offline summarizer pilot uses invented examples with provenance, human review, stable group identifiers, and fixed train, validation, and held-out splits. Its deterministic format, privacy, and leakage checks are pipeline gates, not model-quality scores. A future golden dataset would need independent review and versioning before any model claim.

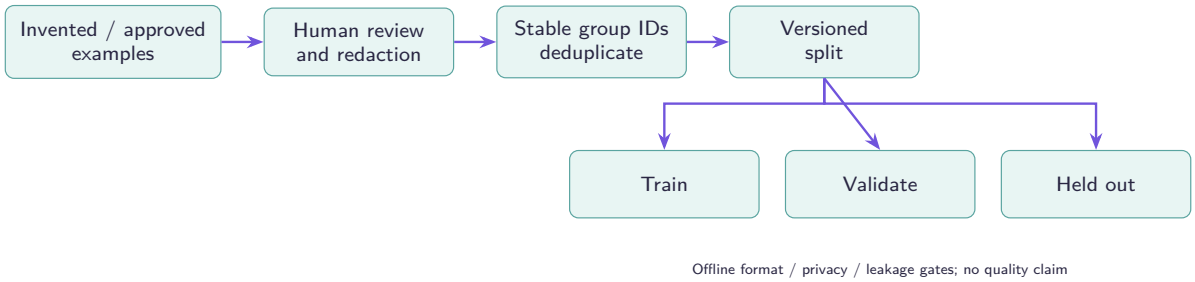


Figure 4. Grouping and review precede evaluation.

6.5 Experiment and deployment control planes

The public refinery architecture proposes isolated experiment jobs and a separate Git-reviewed deployment path. A versioned result, threshold evidence, privacy review, human approval, and rollback reference would be needed before promotion. None of this proposed control plane is deployed by this demo.

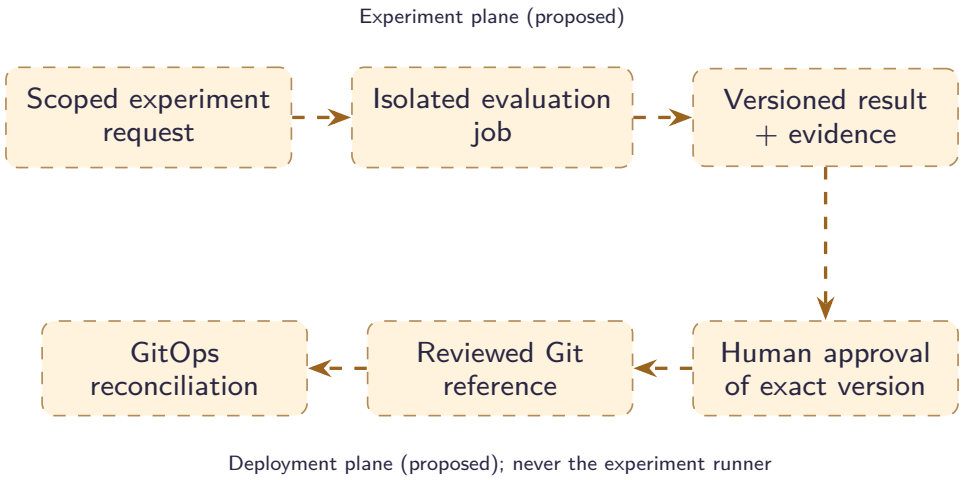


Figure 5. Experiments do not deploy themselves.

6.6 Manually placed compute constellation

For a development environment, assume machines stay on and humans choose app-to-host placement. A simple Compose setup is the first design option; one K3s cluster is another option if its operational need is demonstrated. The refinery remains a separate future layer. The earlier two-cluster availability sketch is superseded for this scenario.

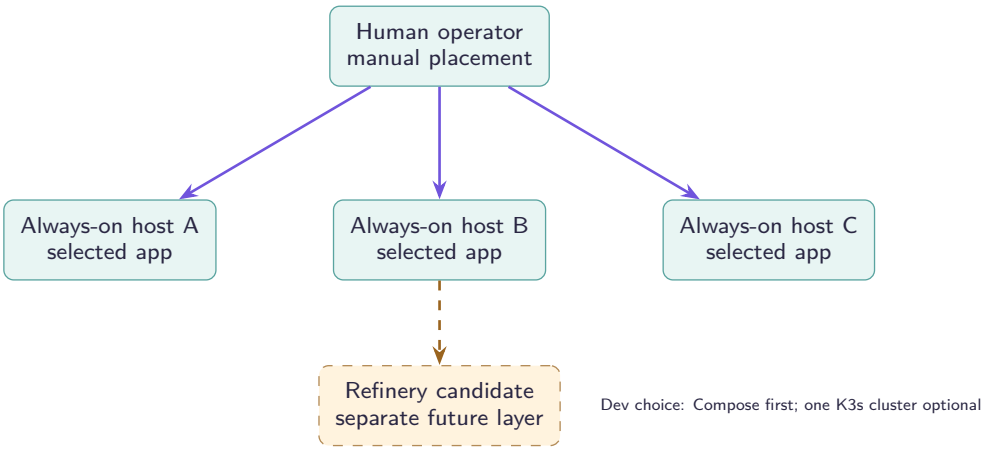


Figure 6. Solid hosts are an assumption; the dashed refinery is proposed.

6.7 Versioned provenance chain

An approved source passes through review and redaction into a dataset version. Candidate artifacts and promotion records are future design stages, linked to exact evidence and a rollback pointer. This graph is a lineage contract, not a claim that a live training pipeline exists.

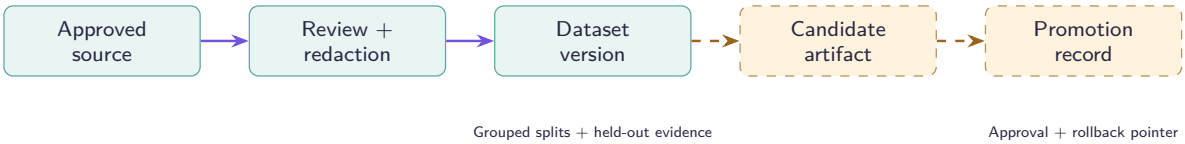


Figure 7. Every promotion should retain its source and review trail.

6.8 Observation versus failure

Process activity ages between observations. A missing update creates uncertainty; inactivity alone is not proof of failure. A verified process check is required before control. This distinction avoids turning stale metadata into an automatic restart trigger.

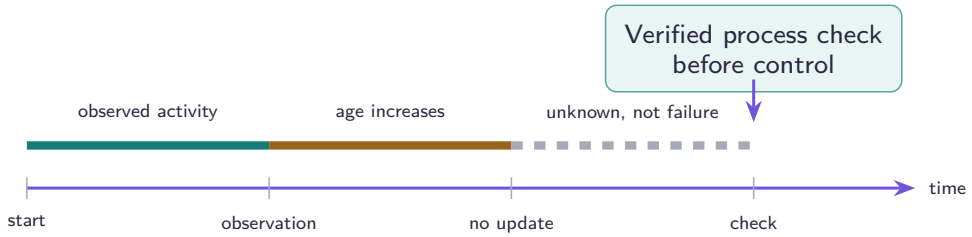


Figure 8. Fresh, aging, and unknown are observation states.

Source boundaries. Public design context: refinery architecture draft and local cloud draft. Operational patterns are abstracted from access-controlled repositories; no private implementation, chat content, identifiers, credentials, or personal records are reproduced here. The eight diagrams are original vector illustrations under this repository's MIT license.